

## Final Project Plan:

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### Sprint 1 — Exploratory Data Analysis (DONE)

Deliverables: 3 EDA notebooks (ACS demographics, GTFS transit, GeoPackage accessibility), Sprint 1 presentation.

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### Sprint 2a — Composite Equity Indicators (DONE)

Deliverables: v3 notebook (512 tracts, 7 indicators, equity priority score), master tract-level CSV, results overview workbook (.xlsx), approach report (.docx).

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### Sprint 2b — Predictive Modeling

*Goal: predict where transit gaps/inequities are likely to occur and what drives them.*

- **2b.1 — GTFS feature engineering.** Compute granular transit service metrics from the 943K stop-time records: headway per stop (avg minutes between arrivals by time-of-day band), service span per stop (first to last trip), frequency by time band per tract (trips/hour at peak, midday, evening, late), weekend vs. weekday service gap per tract, route diversity per tract. Spatially join stops to tracts via lat/lon. These become both predictive features for ML and the adjustable parameters for Sprint 3 simulation.
- **2b.2 — Multi-year ACS data pull.** Pull 2019-2023 ACS 5-Year vintages from Census API for the same tract-level variables. Compute per-tract temporal trends (slope of poverty, rent burden, vehicle access over 5 years).
- **2b.3 — Feature consolidation.** Combine into one modeling dataset: current demographics (from 2a), temporal trend slopes (from 2b.2), GTFS service metrics (from 2b.1), spatial features (neighboring tract scores), and interaction terms (e.g., poverty × transit desert, headway × no-vehicle rate).
- **2b.4 — Regression model (interpretable).** Linear/logistic regression to identify what drives equity priority scores. Output: coefficient table showing which factors matter most and by how much. This is the explainability layer for Deloitte stakeholders.
- **2b.5 — ML model (predictive engine).** Train XGBoost/LightGBM to predict equity scores given tract features + service parameters. Model must accept service inputs (headway, frequency, span, stop count) as features so it can re-predict under changed

conditions. This becomes the simulation backend for Sprint 3.

- **2b.6 — Risk scoring and time series findings.** Use temporal trends + ML predictions to flag tracts trending toward Critical. Identify fragile tracts where demographics are worsening. Document findings.
- **2b.7 — Validation and reporting.** Cross-validation, residual analysis, feature importance plots, model comparison (regression vs. ML). Notebook + summary report.

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## Sprint 3 — Simulation Engine

*Goal: simulate impact of proposed service changes on equity outcomes, powered by the ML model.*

- **3.1 — Define simulation parameters.** Map user-adjustable levers to ML model inputs: headway changes (increase/decrease frequency), service span extension (add evening/late service), stop additions/removals per tract, route additions, weekend service parity. Each lever modifies specific GTFS-derived features from 2b.1.
- **3.2 — Build the simulator.** Takes a proposed service change → modifies the relevant tract-level features → runs ML model to re-predict equity scores → outputs before/after comparison (score deltas, tier shifts, affected population).
- **3.3 — Scenario validation.** Run 3-5 reference scenarios to sanity-check (e.g., "double evening frequency in Critical tracts," "extend service span to midnight in top-20 priority tracts"). Verify outputs behave sensibly. Document.

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## Sprint 4 — Interactive Dashboard

*Goal: visualize transit access and let users run dynamic simulations in real time.*

- **4.1 — Dashboard design.** Layout: choropleth map of equity scores by tract, tier summary panel, tract detail drill-down, simulation control panel (sliders/dropdowns for service parameters), before/after comparison view, trend alerts for at-risk tracts.
- **4.2 — Build core dashboard.** Plotly Dash or Streamlit. Map view with tract coloring by tier/score, filter by neighborhood/tier/indicator, click-to-inspect tract profiles.
- **4.3 — Integrate simulation.** Connect Sprint 3 simulator to dashboard UI. User adjusts service parameters (headway, frequency, span, stops) via controls → ML model

re-predicts → map and scores update in real time. Before/after side-by-side.

- **4.4 — Alerts and recommendations.** Surface at-risk tracts from 2b.6 trend analysis, display feature importance insights from 2b.4 as actionable recommendations, export capability.

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**Deliverable mapping:**

| <b>Deloitte Requirement</b> | <b>Sprint</b> | <b>What It Delivers</b>                                                       |
|-----------------------------|---------------|-------------------------------------------------------------------------------|
| Exploratory Data Analysis   | Sprint 1      | Done                                                                          |
| Prediction Model Results    | Sprint 2b     | Regression (drivers) + ML (predictions) + time series (trends) + risk scoring |
| Simulation Model Results    | Sprint 3      | What-if simulator powered by ML, validated scenarios                          |
| Interactive Dashboard       | Sprint 4      | Real-time map + dynamic simulation UI + alerts + recommendations              |

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